Yazdan Bayat

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LinkedIn | GitHub | Hugging Face

Education

K. N. Toosi University of Technology *Sep 2022 – Jan 2027 (expected)*
B.Sc. in Electrical Engineering
Orientation: Artificial Intelligence & Robotics
GPA: 17.64/20.00

Research Interests

2D & 3D Computer Vision
Health Data Modeling
Generative Models (Diffusion & GAN models)
Image and Video Analysis
Neural Implicit Representation
Robotic Perception & Vision-Based Control

Research Experience

ARAS Laboratory, KNTU *Oct 2023 – Present*
Undergraduate Research Assistant
Cataract Surgery Difficulty Assessment GitHub

- Created, annotated, and organized surgical microscope-image datasets for four-class cataract severity classification and binary poor pupil dilation detection.
- Developed dataset preprocessing and metadata workflows, including video-identifier mapping and duplicate-mapping audits.
- Fine-tuned and compared CNNs (EfficientNet, ResNet, DenseNet, VGG) and transformers (ViT, DeiT, Swin) using transfer learning.
- Evaluated CNNs using video-grouped five-fold cross-validation, keeping frames from each video within one fold; transformer cross-validation is ongoing.
- Applied data augmentation, label smoothing, and differential learning rates; saved checkpoints, training logs, and evaluation outputs for analysis.

Ongoing Paper

Cataract Surgery Difficulty Assessment Using Computer Vision
In progress; working title.
Investigating cataract severity and poor pupil dilation using microscope images and deep learning.

Teaching Experience

Teaching Assistant – Linear Control Systems, Prof. Hamid D. Taghirad *Sep 2025 – Present*
Teaching Assistant – Communication Systems, Assoc. Prof. Mohammad Ali Sebt *Sep 2025 – Present*
Teaching Assistant – Differential Equations, Assoc. Prof. Hadi Aliakbarian *Jan 2025 – May 2025*
Teaching Assistant – Electromagnetics, Dr. Zahra Ghatan Kashani *Jan 2025 – Present*

Selected Courses

Introduction to Artificial Intelligence
Robotics and Machine Vision
Database Systems
Fundamentals of Intelligent Systems
Introduction to Machine Learning
Modern Control Systems

Selected Projects

Isolated Sign-Language Recognition on WLASL GitHub

- Collaborative project comparing RGB-based Swin Transformer and skeleton-based ConvBiLSTM approaches to isolated sign-language recognition.
- Explored landmark-based sequence representations; project materials include training notebooks, validation logs, reports, and a demonstration application.

ROS Canny Edge Detection GitHub

- Developed a ROS 1 Noetic node for processing camera images with OpenCV-based Canny edge detection.
- Integrated live parameter tuning through dynamic reconfigure and visualized camera images and edge outputs in RViz.

Hand-Gesture Recognition GitHub

- Trained and evaluated an LSTM model for temporal hand-gesture classification.

Additional Coursework Projects: MNIST Classification; CIFAR-10 Classification; Titanic Survival Prediction; Graduate Admission Modeling; Traffic-Light Controller Simulation.

Honors & Awards

- Placed in the top 1 % among more than 145,000 participants in the Iranian Nationwide University Entrance Exam (Concours)
- Ranked 46 among 12,000 nationwide participants in the Iranian Physics Olympiad

Language

Persian: Native proficiency
English: Professional Proficiency (TOEFL iBT: 102 – Reading 25, Listening 26, Speaking 24, Writing 27)

References

Prof. Hamid D. Taghirad
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